

Annex 6

World Health Organization/United Nations Population Fund specifications for personal lubricants

Replaces World Health Organization/United Nations Population Fund specifications for plain lubricants, WHO Technical Report Series No 1025, 2020, Annex 11.

For any further information or request, please send an email to the Norms and Standards for Pharmaceuticals (NSP) Team at WHO, at email nsp@who.int.

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Background

The *World Health Organization/United Nations Population Fund specifications for plain lubricants* were recommended for adoption at the fifty-fourth meeting of the World Health Organization (WHO) Expert Committee on Specifications for Pharmaceutical Preparations held in 2019 and published as Annex 11 in WHO Technical Report Series No. 1025, 2020 ([1](#)).

These specifications were developed following a detailed technical review conducted at the United Nations Population Fund (UNFPA) Global Consultation on Personal Lubricants in November 2016 in Bangkok, Thailand, and a follow-up meeting, primarily with lubricant manufacturers, held in conjunction with the thirty-fourth ISO/TC 157 (International Organization for Standardization, Technical Committee 157 for Non-Systemic Contraceptives and STI Barrier Prophylactics) meeting in George Town, Penang, Malaysia, in September 2017.

The Global Consultation on Personal Lubricants was convened to review the safety of personal lubricants, as research had shown that users may experience irritation, burning and damaging effects to vaginal and rectal tissue, and to examine ways to produce, procure and distribute safer products for all. Hosted by UNFPA, the United States Agency for International Development (USAID), WHO, and the International Planned Parenthood Federation (IPPF), the meeting brought together more than 80 manufacturers, researchers and technical experts, sexual health advocates and educators, and international organizations that procure lubricants for governments or local organizations.

The status of the WHO/UNFPA/FHI360 (Family Health International) advisory note on *Use and procurement of additional lubricants for male and female condoms*, published in 2012 ([2](#)), was also reviewed at the Global Consultation. It was agreed that the majority of the recommendations made in that note were still valid, and they have been incorporated in this specification. The recommendation that polyquaternium compounds should be avoided was found to be no longer supportable and has not been included in this specification.

Based on further needs and to align with the latest product and quality requirements, UNFPA, in consultation with stakeholders, proposes the revision of Annex 11 (WHO Technical Report Series No. 1025, 2020).

1. Introduction

These World Health Organization/United Nations Population Fund (WHO/UNFPA) specifications establish quality assurance and regulatory compliance standards for personal lubricants. These specifications are designed to ensure that lubricants meet stringent safety, efficacy and quality standards, safeguarding users from potential adverse effects. The document outlines the requirements for

water-based and silicone-based lubricants, focusing on their physical properties, ingredient safety, compatibility with condoms, microbiological standards, biocompatibility requirements, shelf-life studies, packaging, labelling and lot-by-lot testing. The document does not cover lubricants with added fragrances, colours, spermicides, herbal ingredients or special ingredients claiming specific pleasure-enhancing properties. It also excludes lubricants intended for uses other than with male and female condoms in reproductive health programmes.

Key updates in these specifications are as follows.

- The water-based lubricant viscosity specifications in this document are based on data submitted by manufacturers to UNFPA.
- The pH specification has been revised.
- Environmental control requirements for manufacturing lubricants have been included.
- Manufacturers shall confirm that the preservative system remains effective throughout the shelf-life of the product. The maximum period of use after first opening of multidose containers shall be specified and confirmed by testing.
- Validated microbiological test methods shall be used for testing to confirm they are effective in the presence of the preservatives used in the formulation.
- Manufacturers should also ensure the absence of the pathogen *Burkholderia cepacia* complex.
- Clarification on the method for calculating osmolality from diluted lubricant samples has been included.
- Purified water used in manufacturing water-based lubricants must meet pharmacopoeial standards.

This document is intended for use by lubricant manufacturers, procurement agencies, regulatory authorities and quality assurance personnel involved in the production, procurement and distribution of lubricants for reproductive health programmes. It is also relevant to international organizations, nongovernmental organizations and other stakeholders involved in reproductive health and family planning initiatives.

2. Requirements

Manufacturers shall include in their product dossier evidence to confirm that the lubricant complies with the requirements listed in [Table A6.1](#). Verification of conformance to these requirements is assessed by review of the product dossier.

2.1 **Generic requirements**

Table A6.1
Generic requirements and descriptions

Requirements	Description
Definition and general properties	Description Water-based and silicone-based lubricants shall be clear, translucent or white gels or viscous liquids. They shall be free from lumps, foreign matter and particulate contamination. Water-based lubricants shall be water washable and non-staining. Silicone-based lubricants shall be non-staining.
	Ingredients Lubricants shall contain only ingredients that are safe for human use in contact with vaginal mucosa and skin during sexual intercourse. The ingredients shall be non-irritant and nontoxic and shall not liberate any toxic or harmful substance during storage and use. Lubricants shall be free from added fragrance, colour, spermicides, herbal ingredients and special ingredients that claim specific pleasure-enhancing properties. Purified water used in the manufacture of water-based lubricants shall meet recognized pharmacopoeial standards, such as those of the <i>United States Pharmacopeia</i> , <i>European Pharmacopoeia</i> or <i>The International Pharmacopoeia</i> . Manufacturers may also use deionized water with bioburden level less than 100 colony-forming units per millilitre (CFU/mL). Silicone lubricants shall contain a minimum of 30% polydimethylsiloxane (dimethicone), with a viscosity of 5 centipoise (cPs) and above (mixtures of polydimethylsiloxanes with different viscosities are permitted).
	Compatibility with condoms Lubricants shall be compatible with male and female condoms (any exceptions shall be noted in the labelling). Testing shall be conducted according to ASTM D7661 (3) or ISO 19671:2018 (4) and ISO/TS 23148:2024 (5). When testing silicone lubricants containing volatile cyclomethicone, the conditioning of the condoms in the presence of the lubricants should be done under occlusive conditions to prevent evaporative loss of the cyclomethicone.

Table A6.1 *continued*

Requirements	Description
Definition and general properties <i>continued</i>	Preservatives Water-based lubricants shall contain suitable preservatives to protect against microbial contamination. The lubricant shall be manufactured under conditions that maintain control of bioburden. The effectiveness of the preservative system shall be confirmed by testing, for example, using <i>United States Pharmacopeia</i> <51> on antimicrobial effectiveness testing (6) or <i>European Pharmacopoeia</i> 7.0 (5.1.3) on efficacy of antimicrobial preservation (7). Manufacturers shall confirm that the preservative system remains effective throughout the shelf-life of the product. The maximum period of use after first opening of multidose containers shall be specified and confirmed by testing. Sterility Lubricants may be supplied sterile in unit dose containers. Manufacture Lubricant shall be manufactured in accordance with certified quality management systems and in compliance with national and regional regulatory requirements. The quality management system shall comply with ISO 13485:2016 (8). Lubricant shall have regulatory approval, such as: • European Union CE mark (9) or United States Food and Drug Administration 510(k) clearance (10); or • lubricants that have a market clearance issued by an International Medical Device Regulators Forum management committee member. Manufacturers are advised to follow the guidance given in WHO Technical Report Series No. 1010 (2018), Annex 8, <i>Guidelines on heating, ventilation and air-conditioning systems for non-sterile pharmaceutical products</i> (11), along with its interpretation in WHO Technical Report Series No. 1019 (2019), Annex 2 (12). The heating, ventilation and air-conditioning system and classification of cleanrooms in production areas should be appropriately designed and operated based on the assessment of the risk of potential contamination of the product. It is strongly recommended that positive air pressures are maintained in the chemical handling, mixing and primary packaging areas to prevent ingress of any contamination. Appropriate validated cleaning and sanitization procedures shall be implemented for all equipment, manufacturing areas and primary packaging materials to prevent contamination.

Table A6.1 *continued*

Requirements	Description
Definition and general properties <i>continued</i>	Lubricity There are currently no specification requirements for lubricity, nor are there any recommended methods for measuring lubricity. Manufacturers who specify lubricity requirements should submit details of the specification and test method to UNFPA. Similarly, manufacturers who test for the retention of lubricity over the time of use should submit details of the test method and requirement. Viscosity There are currently insufficient published data to define an optimal viscosity range for water-based lubricants. The viscosity specifications in this specification are based on data submitted by manufacturers to UNFPA. Water-based lubricants usually exhibit shear thinning viscosity behaviour, the measured viscosity being highly dependent on the conditions used to make the measurement. For this reason, viscosities specified in this specification for water-based lubricants are based on measurements made at 25 ($\pm$ 2) °C using a Brookfield or equivalent rotational viscometer with spindle number LV-4 at a rotation speed of 60 revolutions per minute (rpm).
Composition	The manufacturer shall submit to procurement agencies full composition details of the lubricant, with the quantities and specifications of individual ingredients used. Where applicable, ingredients shall conform to relevant pharmacopoeia standards (for example, <i>United States Pharmacopoeia</i>, <i>European Pharmacopoeia</i>, <i>The International Pharmacopoeia</i>). For proprietary or trade name ingredients, the corresponding material safety data sheets and technical specifications shall be submitted. Water-based lubricants shall be formulated to comply with the requirements listed below. • Osmolality shall be less than 1200 milliosmoles per kilogram (mOsm/kg) of water. This osmolality limit can be achieved by keeping the total glycol content below about 8.3 mass fraction (%w/w).⁹

⁹ This limit may be varied depending on the specific glycols used.

Table A6.1 *continued*

Requirements	Description
Composition <i>continued</i>	• If it is necessary to dilute the lubricant to measure the osmolality, an estimate of the undiluted osmolality shall be made by multiplying the measured viscosity by a dilution factor based on the water content of the lubricant before and after dilution and not on the ratio of lubricant volumes before and after dilution. • The pH of the lubricant shall be in the range of 4.0 to 7.0, with an allowable tolerance of ± 1.0.¹⁰ The pH of the vagina is typically around 4.0. The old limit effectively restricted manufacturers to targeting a pH of not less than about 4.5 in order to remain within the specification. The new limits allow manufacturers to target the normal vaginal pH of about 4.0 without going outside the specification. The initial viscosity of water-based lubricants shall be specified by the manufacturer in the range 5000 to 20,000 cPs. A tolerance of $\pm 15\%$ shall be applied to the specified value. The viscosity shall be measured using a Brookfield or equivalent rotational viscometer at $25 (\pm 2) ^\circ\text{C}$ fitted with spindle number LV-4 at a rotation speed of 60 rpm. The initial viscosity of silicone lubricants shall be specified by the manufacturer along with the method and conditions of measurement. A tolerance of $\pm 15\%$ shall be applied to the specified value. The viscosity of both water-based and silicone-based lubricants shall not decrease by more than 30% from the initial value over the product's shelf-life period. Accelerated stability studies may be used to confirm this requirement.
Biocompatibility	Lubricants shall be evaluated for biocompatibility risks according to the requirements of ISO 109931:2025 (13). As a minimum, the evaluations shall include cytotoxicity according to ISO 10993-5:2009 (14), skin sensitization according to ISO 10993-10:2021 (15) and skin irritation according to ISO 10993-23:2021 (16). Depending on national or regional regulatory requirements and whether any changes have been made to the formulation or manufacturing process, additional evaluations specified in other parts of ISO 10993-1:2025 might be necessary. These include systemic toxicity, local effects after tissue exposure and genotoxicity.

¹⁰ Lubricants with a low buffering capacity that do not disturb the pH of the vagina or rectum are preferred.

Table A6.1 *continued*

Requirements	Description
Biocompatibility <i>continued</i>	The toxicity study reports shall be reviewed and interpreted by a qualified toxicologist or other suitably qualified expert. Full reports of biocompatibility assessments shall be submitted as part of the product dossier.
Bioburden levels	Lubricants need not be sterile. However, they shall be subjected to control of microbial contamination by appropriate measures taken in formulation, manufacturing and packing operations. In the finished product, bioburden levels (total aerobic microbial count) shall be maintained below 100 CFUs per gram, and total combined yeast and mould count shall be maintained below 10 CFUs per gram – see <i>United States Pharmacopeia</i> <1111> (17), <i>European Pharmacopoeia</i> (18) and <i>Japanese Pharmacopoeia</i> (19). There shall be an absence of <i>Pseudomonas aeruginosa</i>, <i>Staphylococcus aureus</i>, <i>Candida albicans</i>, <i>Escherichia coli</i> and <i>Burkholderia cepacia</i> complex. These requirements apply to both water-based and siliconebased lubricants. Bioburden levels shall be maintained at the above levels during storage and repeated opening of a container during multiple use. Lubricants shall comply with the evaluation of preservative efficacy, performed as per the requirements of a relevant pharmacopoeia. All microbiological test methods shall be validated to confirm they are effective in the presence of the preservatives used in the formulation. If the lubricant is supplied sterile in unit dose containers, the sterility assurance level shall be 10^{-6}.
Shelf-life and stability	Lubricants shall have a minimum shelf-life of three years from the date of manufacture when stored under the labelled storage conditions. To ensure compatibility with condom storage recommendations and shelflife estimates, real-time studies shall be conducted within the temperature range of 28 °C to 35 °C. The humidity shall be maintained at 75 (± 5) % relative humidity, to ensure conformity with climate zone IVb (hot, higher humidity) requirements.

Table A6.1 *continued*

Requirements	Description
Shelf-life and stability <i>continued</i>	In line with International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) guidelines (20), accelerated studies shall be conducted at 40 (± 2) °C and 75 (± 5) % relative humidity. Manufacturers may elect to use higher temperatures, such as 50 °C and 60 °C, providing the results can be correlated with realtime shelf-life estimates at 28 °C to 35 °C. Higher temperatures (for example, 50 °C and 60 °C) may be used to support extrapolation, but these studies are not substitutes for real-time stability studies. Bioburden results obtained at higher temperatures may also be misleading and should be treated with care. For waterbased lubricants, manufacturers should include freeze/thaw cycling in their stability studies to confirm that the lubricants can tolerate freezing. Manufacturers should also confirm that osmolality remains within specifications at the end of the stability study and undertake intermediate osmolality measurement if any significant changes occur in the water content or viscosity of the lubricant, and, in the case of lubricants packed in sachets, the weight of the sachets during the course of the study. The following critical parameters shall be monitored throughout the stability studies: • pH • bioburden • viscosity • odour • physical condition (such as clarity or texture) • appearance and colour • packaging integrity. For water-based lubricants, add: • preservative content (assay) • microbiological challenge testing. For silicone-based lubricants containing cyclomethicone: • monitoring of weight loss due to any loss of volatile material through the packaging. Lubricants shall remain within the manufacturer’s specification for the duration of the shelflife period. The data and report on accelerated stability studies and ongoing real-time studies shall be submitted as part of the product dossier. Regular testing intervals (for example, 0, 3, 6, 9, 12, 18, 24 and 36 months) shall be applied.

Table A6.1 *continued*

Requirements	Description
Compatibility with condoms	The manufacturer should submit reports of compatibility studies conducted on the use of lubricant with male and female condoms made from natural rubber latex and synthetic materials. Any exceptions from testing or incompatibilities shall be noted.
Packaging	Individual containers Lubricants shall be packed in hermetically sealed tamperevident containers that facilitate multiple delivery of lubricant. Examples are collapsible or squeeze tubes and containers with a suitable delivery system for application of lubricant. It is recommended that containers should be made of recyclable materials that are compatible with the lubricant, as substantiated by stability studies and shelflife claims. The containers shall not have sharp edges. They shall not liberate any toxic or harmful substance during storage and use of the product. The individual containers shall be free from leakage of lubricant. The recommended nominal contents for multidose containers are 35 g, 50 g and 82 g. Other sizes may be considered, depending upon programme requirements. The recommended nominal contents for a singledose sachet is 3 g for silicone lubricants and 4–5 g for waterbased lubricants. Pack contents are based on the amount of lubricant that can be expressed from the pack under normal use. This will be evaluated by weighing 20 full primary containers individually and weighing them again after squeezing out their contents. Alternatively, the weight of lubricant expressed may be determined directly by collecting it in a tared container or dish. Secondary packing The individual containers shall be packed in secondary distribution packages of an appropriate size as per programme requirements (for example, 25 units per secondary pack). Cardboard boxes shall be marked or certified Forest Stewardship Council (or equivalent). They shall only contain paper or cardboard. Plastic coating shall not be used. Shipper cartons Shipper cartons shall be marked or certified Forest Stewardship Council (or equivalent). They shall be made of a minimum of 40% recycled or postconsumer material.

Table A6.1 *continued*

Requirements	Description
Packaging <i>continued</i>	The shipper carton should only contain paper or cardboard. Plastic coating shall not be used. Plastic carton liners, where used, shall be made from recycled or biodegradable materials. The recommendations relating to packaging in this specification may be varied depending on the intended use of the lubricant. Full details of the required packaging should be agreed in advance and specified in purchase orders.
Labelling	Individual containers Labelling requirements may be subject to local regulatory requirements. Subject to any local requirements, the individual containers shall be marked with the following details: • contents (specify if it is water or siliconebased lubricant) • quantity of lubricant that can be expressed from the container in normal use • if in a multidose container, advice on the amount of lubricant to be used • manufacturer's name and address • batch or lot number • expiry date (in YYYYMM format) • storage conditions – store at an average temperature below 30 °C and avoid exposure to direct sunlight • warnings or special notes, if any • maximum time period in which the contents can be used after the container was first opened • a list of any ingredients that may be an irritant or that could cause allergic reactions • a statement that the lubricant is compatible with male and female condoms (any exceptions, such as male polyurethane condoms, shall be stated on the package) • a statement that the lubricant is not a contraceptive and does not protect against pregnancy, sexually transmitted infections and HIV. To protect against pregnancy and sexually transmitted infections, the lubricant must be used with a condom.

Table A6.1 *continued*

Requirements	Description
Labelling <i>continued</i>	Secondary packaging • contents • quantity • manufacturer's name and address • batch or lot number • date of manufacture and expiry date (in YYYYMM format) • storage conditions • warnings or special notes, if any. Shipper cartons (or as per UNFPA shipping instructions to be provided by the buyer): • UNFPA logo • UNFPA project number • UNFPA purchase order number • country of destination • contents as waterbased lubricants • quantity • manufacturer's name and address • batch or lot number • date of manufacture (in YYYYMM format) • expiry date (in YYYYMM format) • weight • volume • storage conditions text: "Store in wellventilated, dry storage conditions with an average temperature of less than 30 °C away from direct sources of heat, including sunlight" • warnings or special notes, if any, to be defined by the manufacturer • any special shipping instructions defined by the manufacturer • all markings must be clearly legible. <i>Note:</i> symbols used in the labelling, and information to be supplied (for example, an information for use leaflet), shall comply with the current version of ISO 15223 Part 1 (21).

2.2 Lot-by-lot testing requirements

The manufacturer shall submit a certificate of analysis, signed by an authorized quality representative, for each batch or lot of lubricant supplied, confirming compliance with the requirements of this section and referencing the applicable specification version. This section may also be used by accredited or approved laboratories for the independent testing of lubricants ([Table A6.2](#)).

Table A6.2
Lot-by-lot testing requirements

Parameter	Requirements	Verification
Description	Water-based lubricant shall be clear, translucent or white gel or viscous liquid, free from lumps and foreign matter, and water washable. Silicone lubricants shall be clear, translucent or white gels or viscous liquids free from lumps and foreign matter and be nonstaining.	Visual inspection shall be performed on samples weighing about 5 g drawn randomly from five individual containers per lot.
pH	Shall be in the range 4.0 to 7.0 with a tolerance of ± 1.0 .	A composite sample (weighing about 10 g) prepared from five randomly selected individual containers shall be tested.
Initial viscosity water-based lubricants	Shall be within a tolerance of $\pm 15\%$ of the specified viscosity value.	Viscosity shall be measured at $25 (\pm 2) ^\circ\text{C}$ using a Brookfield or equivalent rotational viscometer with spindle number LV-4 at a rotation speed of 60 rpm.
Initial viscosity silicone lubricants	Shall be within a tolerance of $\pm 15\%$ of the specified viscosity value.	Viscosity shall be measured as specified by the manufacturer.

Table A6.2 *continued*

Parameter	Requirements	Verification
Bioburden	Bioburden levels (total aerobic microbial count) shall be maintained below 100 CFUs per gram, and total combined yeast and mould count shall be maintained below 10 CFUs per gram. There shall be an absence of <i>Pseudomonas aeruginosa</i>, <i>Staphylococcus aureus</i>, <i>Candida albicans</i>, <i>Escherichia coli</i> and <i>Burkholderia cepacia</i> complex. Sterility (if claimed) shall be to the sterility assurance level of 10⁻⁶.	Testing as per <i>The International Pharmacopoeia</i> (22), <i>United States Pharmacopeia</i> (23), <i>European Pharmacopoeia</i> (24) or <i>Japanese Pharmacopoeia</i> (25). Recommended testing frequency: • Every lot shall be tested for the first 10 production lots. • If all 10 conform to specification, the testing frequency may be reduced to one in every 10 lots. • If any lot fails, full testing shall resume until 10 consecutive lots again meet specifications.
Packaging and labelling	Shall comply with requirements of packaging and labelling as given in section 2.1, except for material of construction.	Visual inspection shall be performed on a random sample of 13 containers per lot or batch.

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